



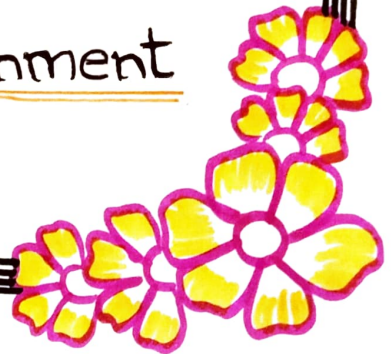
Name :: J. Sravati

Reg no :: 192372141

Course code :: ITAI408

Course :: Ethical Hacking

C01 - AT3 - Assignment



Ethical and practical Disclosure Obligations in a Multi-Party Vulnerability

- ⇒ Cybersecurity professionals have a major responsibility to protect information systems and sensitive data. Vulnerabilities that may affect not only the client tested but also
- ⇒ Third-party software used to many other organizations. Such as situations require careful ethical judgement, legal awareness and professional responsibility.
- ⇒ The situation creates ethical and practical challenges because the consultant has responsibilities toward multiple parties:
 - The client.
 - The Third-party software vendor.
 - Other organization using the software.
 - The general-public who may be affected if the vulnerability is exploited.

Ethical Responsibilities

1. Duty to protect the client

The first responsibility is toward the client hired the consultant.

⇒ The consultant should:

- Inform the client immediately
- Explain the severity of the vulnerability
- Recommend temporary security measures until a permanent fix becomes available.

2. Maintain Confidentiality

Cybersecurity professionals often sign Non-Disclosure agreements [NDA's]

Therefore, they should:

- Keep client information confidential.
- Never disclose sensitive client data publicly.
- Avoid revealing the client's identity without permission.
- Share only the technical details necessary to fix the problem.

3. Avoid causing Harm

One of the most important ethical principles in cyber security is:

The consultant must ensure that:

- The vulnerability is not exploited.
- Attackers do not gain access to the information.
- No unnecessary panic is created.

4. Professional Integrity

Cybersecurity professionals should:

- Act honestly.
- Report accurate findings.
- Avoid exaggerating the severity.
- Never exploit vulnerabilities for personal benefit.

Practical Disclosure Responsibilities

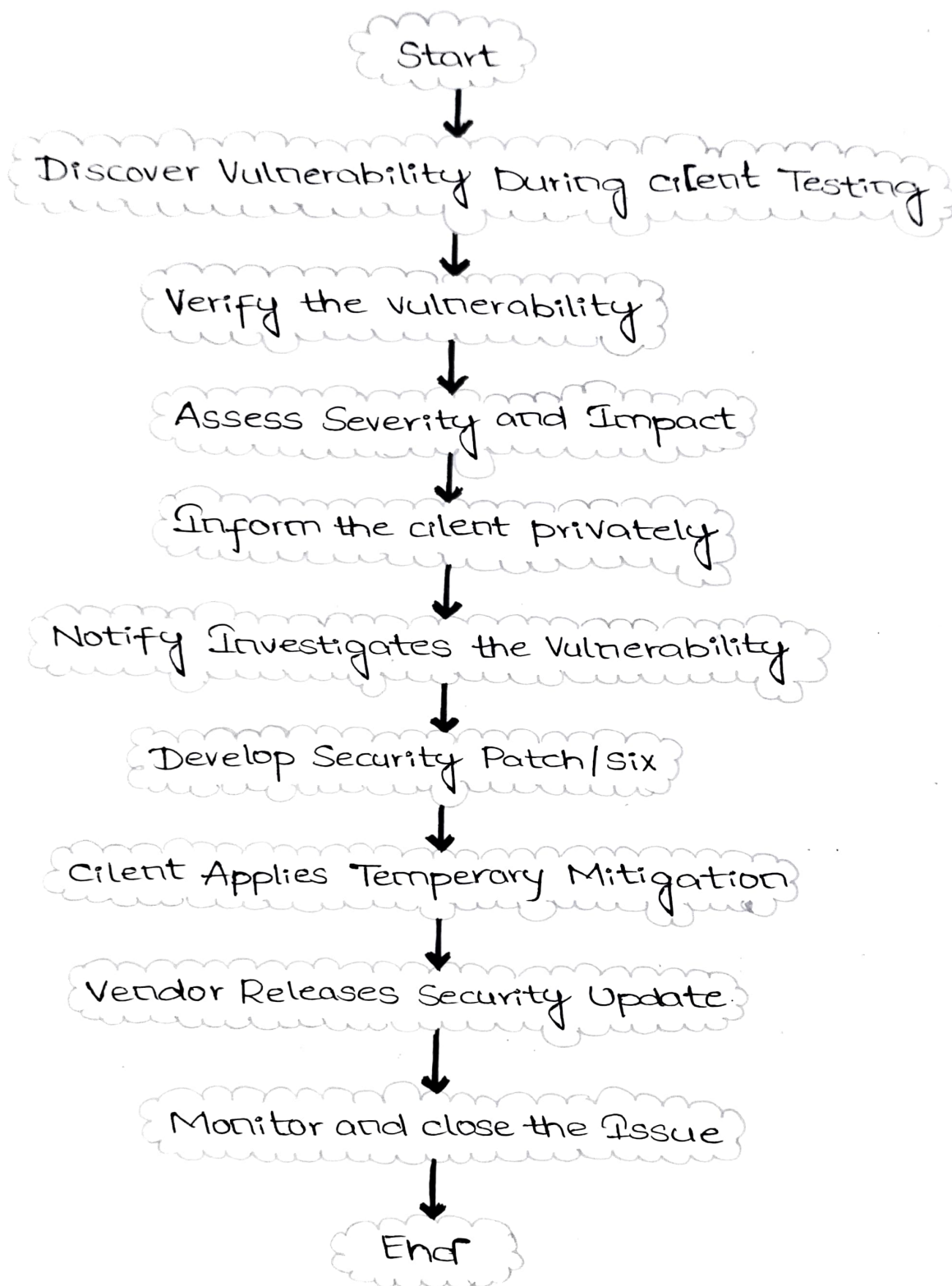
1. Verify the vulnerability

- ⇒ The vulnerability is real.
- ⇒ It can be reproduced.
- ⇒ It is not caused by incorrect system configuration.
- ⇒ The issue exists in the third-party software.

2. Inform the client first

- ⇒ Description of the vulnerability.
- ⇒ Severity level.
- ⇒ Potential impact.

- Immediate Mitigation Strategies.
- Recommended next steps.



Responsibilities Toward Other Organizations

- Thousands of organizations may use the Variable a

of vulnerable software

- ⇒ Although they are not direct clients, they are also risk.
- ⇒ The consultant has an ethical responsibility to reduce a help the risk

- Supporting vendor security advisories.
- Encouraging rapid patch development.
- Helping affected organizations understand the vulnerability.
- Sharing mitigation recommendations after the disclosure.

Legal Considerations

Cybersecurity professionals must obey applicable laws.

- ⇒ Do not access system beyond authorized testing.
- ⇒ Respect software licensing agreements.
- ⇒ Follow contractual obligations.
- ⇒ Avoid unauthorized data collection.
- ⇒ Protect privacy regulations such as
 - GDPR
 - Other local data
 - Protection laws

Coordinated vulnerability Disclosure [CVD]

Typical Steps include:

Step '1' -

Discover the vulnerability

Step '2' -

Verify the issue

Step '3' -

Notify the issue

Step 4 -

Notify the software vendor privately

Step 5 -

Vendor investigates.

Step 6 -

Vendor develops a security patch.

Step 7 -

Organizations install updates.

Step 8 -

Public disclosure occur after users have had time to update.

This process balances transparency with user safety.

Challenges in Multi-party disclosure

Several practical challenges may rise:

⇒ Vendor ignores the report

Some vendors may not respond quickly.

⇒ Slow patch development

Complex vulnerabilities may take months to fix.

⇒ Large number of affected users

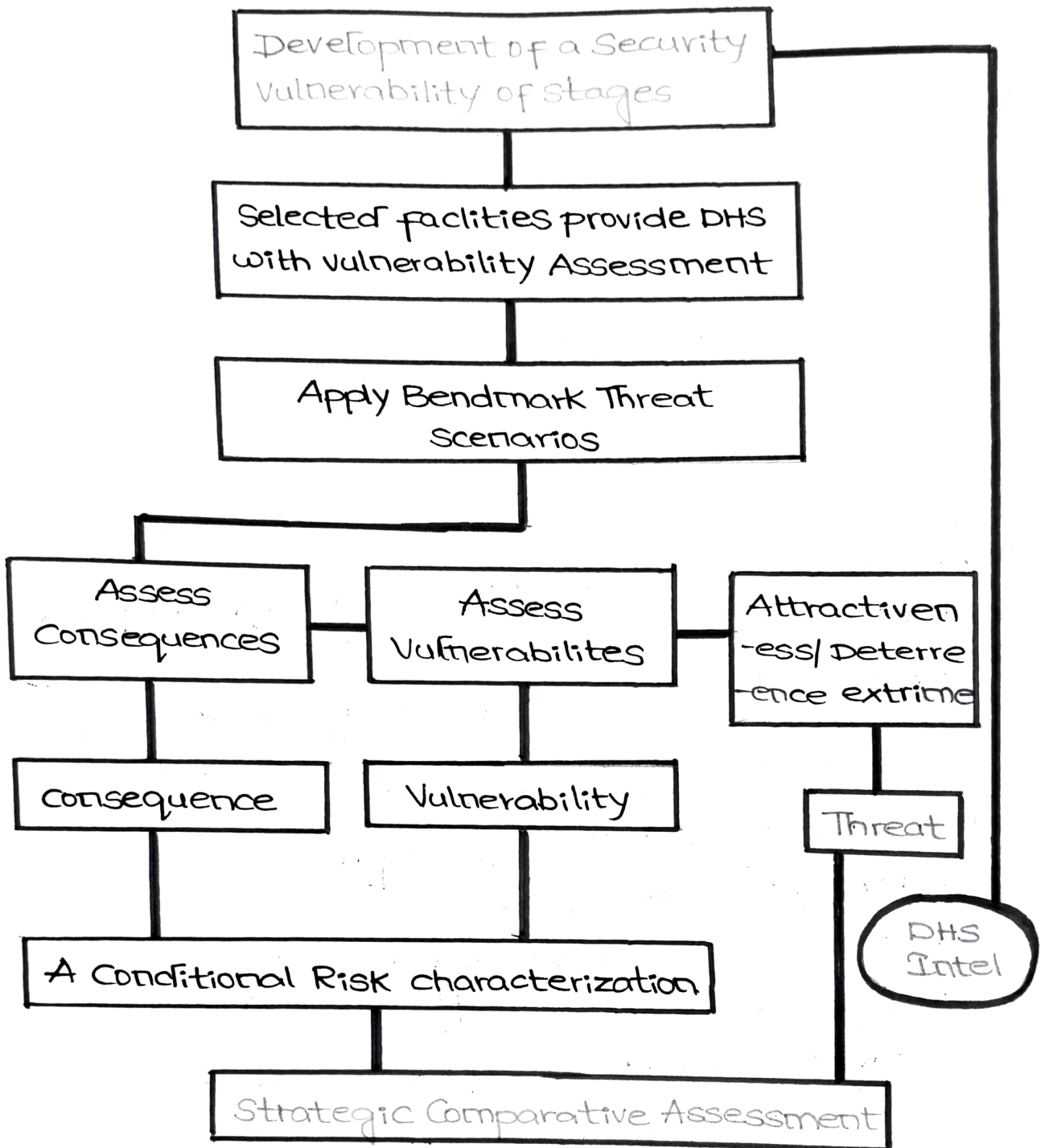
Thousands of organizations may remain vulnerable.

⇒ Risk of public leaks

If information leaks early, attackers may exploit a vulnerability before a fix is available.

⇒ Communication difficulties.

Coordinating among the client, vendor, CERTs.



Best practices -

A cybersecurity professional should:

- Document every finding carefully
- Maintain confidentiality.
- Verify technical evidence.
- Inform the client immediately.

- Contact the software vendor responsibly.
- follow coordinated vulnerability disclosure principles.
- Recommend temporary mitigation measures.
- Avoid public disclosure before a patch is available.
- keep detailed records of communications.
- Continue supporting the client until the issue is resolved.

Conclusion -

- ⇒ Discovering a vulnerability in widely used third-party software creates responsibility that extends beyond a single client.
- ⇒ A cybersecurity professional must protect the client's confidentiality.
- ⇒ Ethical behaviour requires honesty, integrity and a commitment to minimizing harm.
- ⇒ By following coordinated vulnerability disclosure, cybersecurity professionals strengthen trust, and their tools can be exploited on a large scale.
- Discover vulnerability
- Verify the issue
- Assess risk
- Inform client
- Notify software vendor
- Vendor creates and releases a patch.
- Client applies mitigation